

Flood Guard 360

INDIVIDUAL THESIS

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Individual component: Prediction based flood map

Group ID: r26-it-031

Report submitted in partial fulfilment of the requirements for the

BSc(Hons) in information technology

Specialized in information technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Flooding is a recurring hazard in Sri Lanka, and effective response requires information about where flood impacts may be most severe. Flood maps based only on static geographic layers may not reflect changing environmental conditions, while disconnected prediction and response tools can make it difficult for authorities to interpret risk in a practical spatial context. This individual thesis focuses on the prediction-based flood-map component of Flood Guard 360, an integrated AI-, Internet of Things (IoT)- and Geographic Information System (GIS)-based flood-disaster management platform. The component is designed to estimate location-level flood risk from environmental and spatial conditioning factors, including rainfall, river-water level, elevation, land use, drainage and soil-related information where available. The methodology described for the wider project evaluates Random Forest and Gradient Boosting classifiers against a Logistic Regression baseline. Predicted categories—Low, Moderate, High and Critical—are represented as geospatial features and visualized through an interactive Leaflet-based map. The wider research paper reports accuracy and F1-score values of 1.00 for Random Forest and Gradient Boosting and an accuracy of 0.93 with an AUC of 0.95 for Logistic Regression on the stated held-out evaluation; however, the flood-risk test set contains only 40 observations, so these results require cautious interpretation and do not establish general performance across Sri Lanka. The individual contribution is the design and implementation focus on converting model predictions into a location-aware map that can support disaster-management decisions and share risk outputs with other platform modules. Further validation with geographically diverse, independently labelled flood observations and verified spatial datasets is required before operational deployment.

Keywords: flood-risk mapping, GIS, Random Forest, GeoJSON, disaster management

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Mr. Deemantha Siriwardhana, for his guidance and academic support during this individual research and development work. I also acknowledge the support of the Flood Guard 360 research team and the academic staff of the Sri Lanka Institute of Information Technology. Their collaboration and feedback contributed to the broader integrated flood-management project within which the prediction-based flood-map component was developed. I remain responsible for verifying the accuracy of the content, implementation details, data, results and references presented in this thesis.

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LIST OF ABBREVIATIONS

- AI Artificial Intelligence
- API Application Programming Interface
- AUC Area Under the Curve
- CSV Comma-Separated Values
- F1-score F1 Score (Harmonic Mean of Precision and Recall)
- FN False Negative
- FP False Positive
- GeoJSON Geographic JavaScript Object Notation
- GIS Geographic Information System
- GPS Global Positioning System
- IoT Internet of Things
- JSON JavaScript Object Notation
- ML Machine Learning
- ROC Receiver Operating Characteristic
- RF Random Forest
- UI User Interface
- URL Uniform Resource Locator
- UX User Experience
- TN True Negative
- TP True Positive

1 INTRODUCTION

1.1 Overall project context

Flooding can cause damage to homes, roads, public infrastructure and livelihoods, and can disrupt access to rescue services and essential supplies. Flood management therefore benefits from timely information that combines environmental observations with geographic context. Flood Guard 360 is an integrated platform that brings together GIS-based flood-risk prediction and mapping, IoT-based river and drainage monitoring, post-flood health-risk screening, and rescue and relief prioritization. Its purpose is to support coordinated decision-making across different stages of a flood event rather than treating each activity as an isolated system.

1.2 Individual component context

This thesis addresses the prediction-based flood-map component. Its role is to convert environmental and spatial information into risk categories associated with geographic locations and to present those results on an interactive map. The wider design identifies rainfall, river-water level, elevation, land use, drainage and soil-related variables as relevant conditioning factors. The map is intended to let users examine the spatial distribution of predicted risk and relate it to geographic features such as district boundaries, rivers and monitoring points. Risk outputs can also be shared with other Flood Guard 360 modules.

1.3 Problem statement

Flood-related decisions require an understandable view of which locations may face greater risk. A model output presented only as a numerical label or table does not show the geographic distribution of risk, while a static map may not be readily updated when new rainfall or river-level observations become available. The individual problem is how to connect flood-risk classification with geospatial processing and interactive visualization so that location-level predictions can be inspected and used as decision-support information.

1.4 Research gap

Machine-learning classifiers and GIS mapping techniques are established approaches for flood-susceptibility assessment. The gap addressed by this component is the practical connection between prediction and an interactive, location-aware interface within a broader disaster-management workflow. The map must preserve the geographic meaning of predictions, present risk classes consistently, and make outputs available to connected monitoring and response modules. The wider platform seeks to reduce the separation between risk mapping and operational response; this individual thesis concentrates on the prediction-to-map pathway.

1.5 Aim and objectives

The aim is to develop a GIS-based flood-risk mapping component that presents model-generated flood-risk categories in an interactive map and supports information exchange with the wider Flood Guard 360 platform.

The objectives are to: (1) identify and prepare relevant spatial and environmental inputs; (2) support location-level flood-risk classification using the models described in the project research; (3) represent predictions in a GIS-compatible format such as GeoJSON; (4) visualize risk categories through an interactive Leaflet-based web map; (5) design the output so it can be consumed by the platform integration layer; and (6) identify validation requirements and limitations associated with data quality, model performance and geographic generalization.

1.6 Research questions or hypotheses

RQ1: How can environmental and spatial conditioning factors be organized to support location-level flood-risk classification?

RQ2: How can predicted risk categories be converted into a geospatial output and presented clearly through an interactive map?

RQ3: How can flood-map outputs be made available to monitoring, rescue and relief components within an integrated disaster-management platform?

RQ4: What limitations affect the reliability and generalizability of the prediction-based flood map?

This thesis is exploratory and implementation-oriented. It does not claim a new statistical hypothesis beyond the model evaluation reported in the wider project paper.

1.7 Scope limitations and assumptions

The scope is limited to the prediction-based flood-map component, including spatial input preparation, model output representation, interactive visualization and integration requirements. The wider project paper describes Random Forest, Gradient Boosting and Logistic Regression, with risk classes Low, Moderate, High and Critical. The quality of the map depends on the availability, spatial resolution, currency and accuracy of the input layers and labelled flood records. The reported flood-risk test set contains 40 observations, which limits confidence in generalization. The map is a decision-support aid and is not a guarantee that flooding will occur. Live operational performance and field accuracy require further independent validation.

1.8 Individual contribution

The individual component addressed in this thesis is the prediction-based flood map. The contribution is focused on the workflow that links environmental and geographic information to risk classes, converts predicted results into a GIS-compatible representation, and presents them through an interactive map. It also considers how the risk layer can be shared with the wider Flood Guard 360 platform. Team-level functions such as IoT drainage sensing, health screening and relief prioritization are described only to explain integration context; they are not claimed as individual contributions of this thesis.

1.9 Thesis structure

Chapter 1 introduces the project context, the individual flood-map problem, research gap, aim and scope. Chapter 2 reviews flood-risk mapping, GIS and machine-learning concepts and related approaches. Chapter 3 explains the research design, study materials, workflow and evaluation measures. Chapter 4 presents the component requirements and design. Chapter 5 describes implementation, integration and testing considerations. Chapter 6 reports the results documented in the wider research paper and discusses product-level evidence. Chapter 7 interprets

the findings and limitations. Chapter 8 discusses commercialization and sustainability. Chapter 9 concludes the thesis and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review scope focuses on flood-susceptibility mapping, GIS-based visualization, machine-learning classification and the integration of environmental monitoring with flood-risk information. The supplied project research paper provides the principal project-specific basis for the selected models, risk categories, system architecture and reported evaluation. Relevant foundational and related works listed in the project paper are used to frame the design. This is a focused narrative review rather than a claim of a systematic review with exhaustive database searches or formal screening statistics.

2.2 Domain and theoretical foundations

Flood-risk mapping estimates relative risk or susceptibility for geographic locations using conditioning factors. GIS supports the storage, analysis and visualization of spatial data, while machine learning can model relationships between input variables and labelled risk outcomes. Random Forest combines predictions from multiple decision trees and can represent nonlinear interactions; Gradient Boosting builds an ensemble sequentially to reduce prediction errors; Logistic Regression provides a comparatively simple baseline classifier. GeoJSON allows geographic features and associated attributes to be exchanged between a prediction service and a web map. Leaflet and React-Leaflet support interactive browser-based maps. These techniques provide the conceptual basis for presenting model outputs as spatially interpretable risk categories.

2.3 Related research

The project paper cites research on flood-susceptibility mapping that combines machine learning with GIS, including work by Demissie and colleagues [4]. Breiman's Random Forest work [3] provides the foundation for the ensemble classifier, while Friedman [3] describes gradient boosting. Research on river monitoring and flood monitoring using IoT and sensors [2], [3] provides context for linking map outputs to changing environmental observations. These works support the use of machine learning and spatial visualization as complementary techniques. The project-specific

contribution is the integration of risk mapping with monitoring and response modules, rather than proposing that these underlying algorithms are new. The project paper cites research on flood-susceptibility mapping that combines machine learning with GIS, including work by Demissie and colleagues [4]. Breiman's Random Forest work provides the foundation for the ensemble classifier, while [3] Friedman describes gradient boosting. Research on river monitoring and flood monitoring using IoT and sensors provides context for linking map outputs to changing environmental observations. These works support the use of machine learning and spatial visualization as complementary techniques. The project-specific contribution is the integration of risk mapping with monitoring and response modules, rather than proposing that these underlying algorithms are new.

2.4 Existing systems products and techniques

Common approaches include static flood-susceptibility maps, GIS applications that display observed or modelled flood information, and machine-learning pipelines that classify locations using environmental variables. Static maps can communicate broad spatial patterns but may not automatically reflect new observations. Sensor-based systems can provide current water-level information but may not independently provide a district-wide spatial risk view. A prediction-based interactive map can connect these capabilities by showing modelled risk in geographic context and accepting updated outputs from connected services. The available project materials do not provide a controlled head-to-head evaluation against named commercial products, so no quantitative superiority over specific products is claimed.

2.5 Gap synthesis and conceptual framework

The identified gap is the need to make flood-risk predictions operationally interpretable and exchangeable. The conceptual workflow is:

(1) collect and prepare environmental and spatial input data;

(2) derive the model feature set;

(3) run the selected classifier;

(4) assign each location a risk category;

(5) serialize the output as a geospatial layer;

(6) display the layer in the interactive map;

(7) publish the resulting risk information to the platform integration layer.

Incoming river-level or rainfall observations may trigger a refresh when supported by the deployed integration workflow. The map remains a decision-support component and should be interpreted alongside source-data quality and other operational information.

3 RESEARCH METHODOLOGY

3.1 Research design

The work follows a design-and-development approach. The component is specified as part of an integrated web platform, and assessed through the model evaluation and functional testing described in the project materials. The research combines geospatial data handling, machine-learning classification, web mapping and integration design. Model performance metrics are reported from the supplied research paper; where individual map-specific testing evidence is not available, this thesis identifies the test cases needed rather than claiming that they were completed.

3.2 Data participants or study materials

The study materials are geospatial and environmental inputs identified in the project design: historical flood records, rainfall, river-water levels, elevation, land use, drainage and soil-related variables where available. Geographic layers may include administrative boundaries and river-network data, while predicted outputs are represented as spatial features. The wider project implementation notes refer to district and river shapefiles, a map input CSV and elevation rasters; data completeness and availability must be checked for each run. No human participants are required for the core map-processing workflow, and the supplied materials do not document a formal user study for this individual component.

3.3 Experimental or study procedure

The planned workflow is to inspect and clean the source data; check coordinate reference systems and geometry validity; align spatial records; derive location-based features such as elevation or distance to river when corresponding layers are available; prepare the model feature matrix; apply the trained classifier; map predictions to the defined risk categories; convert the results to GeoJSON or an equivalent GIS layer; and render the output on the Leaflet map. The wider project compares Random Forest and Gradient Boosting with Logistic Regression. The workflow should preserve the same feature definitions used during model training and inference. The resulting layer can then be passed to the integration layer for use by other modules.

3.4 Measures and evaluation metrics

The wider project paper reports accuracy, precision, recall, F1-score and area under the ROC curve (AUC) for flood-risk classification. These measures evaluate classification performance, while map-specific verification should separately examine spatial alignment, valid geometries, feature completeness, correct class-to-style mapping, and successful rendering. For any future independent evaluation, the test data should be separated spatially or temporally from training data where possible. The current project paper reports a held-out test set of 40 observations; this is small and limits the strength of generalization claims.

3.5 Data analysis

The reported analysis compares the candidate classification models using the metrics listed above. In the supplied project paper, Random Forest and Gradient Boosting each report accuracy and F1-score of 1.00 on the stated test set, while Logistic Regression reports accuracy of 0.93 and AUC of 0.95. These figures are reproduced as project-reported results and are not presented as a new experiment performed for this individual thesis. The small test set means the perfect ensemble scores should be treated cautiously. A fuller analysis should include class-wise precision and recall, a confusion matrix, independent spatial validation and assessment of whether the classes are balanced.

3.6 Ethics privacy security and responsible AI

Flood-risk outputs can influence public safety and resource allocation, so the map should communicate uncertainty and avoid presenting predictions as guarantees. Data provenance, update time and known limitations should be documented. Access to administrative functions should follow the wider platform's authentication and role-based authorization design. Location data and operational records should be protected from unauthorized access. Model outputs should support, not replace, decisions by disaster-management personnel, and any public warning should follow authorized procedures. The map should be tested for misleading visual encodings, missing data and stale layers.

3.7 Reproducibility and research artefacts

Reproducibility requires recording the source datasets and versions, coordinate reference systems, feature definitions, preprocessing steps, model version, model parameters, class mapping, and output-generation procedure. The project materials identify Python-based geospatial and machine-learning processing, GeoJSON output and a Leaflet-based frontend. Where possible, code, configuration and test fixtures should be version-controlled. Large spatial datasets may need separate documentation of provenance and licensing. The supplied materials do not provide a complete immutable experiment package, so full reproducibility remains a future improvement.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements: accept the required spatial and environmental inputs; validate and prepare the data; generate risk predictions using the selected model; assign Low, Moderate, High or Critical categories; export predictions as a GIS-compatible layer; render the layer on an interactive map; allow users to inspect the geographic distribution of risk; and make risk outputs available to the platform integration layer. Non-functional requirements include clear visual encoding, acceptable response time for the intended dataset size, data integrity, maintainability, compatibility with the wider web application, secure access to administrative features, and graceful handling of missing or invalid spatial data.

4.2 Component architecture within the complete solution

The component follows data-processing and presentation architecture. Spatial and environmental datasets feed a preparation and feature-engineering stage. The prepared feature set is passed to a Python machine-learning service that produces a risk category for each location. The predictions are serialized into GeoJSON and consumed by the React/Leaflet frontend for map rendering. The wider platform uses Node.js and Express REST APIs, WebSocket/Socket.IO communication and shared storage to exchange information. Risk-map outputs can be consumed by monitoring and rescue/relief workflows. The map component is modular so model inference, and visualization can be tested separately.

4.3 Design alternatives and decisions

A static map image is simple to publish but offers limited interaction and is difficult to update programmatically. A table of predicted classes is easy to generate but lacks spatial context. A browser-based GIS map was selected in the project design because it enables users to explore predicted risk alongside geographic reference layers. GeoJSON provides a straightforward transfer format for geographic features and attributes. Random Forest and Gradient Boosting are included as ensemble candidates, while Logistic Regression provides a baseline. These choices align with

the wider project's stated architecture; the available evidence does not establish that they are universally optimal for all flood regions.

4.4 Detailed design

The input stage checks that records contain valid locations and the features expected by the model. Spatial layers should use compatible coordinate reference systems, and raster-derived features should be sampled at the correct location. The prediction stage returns a class label for each input location. The output stage associates the label with the relevant geometry and properties, then encodes the result in GeoJSON. The presentation stage assigns consistent visual styles to Low, Moderate, High and Critical classes and displays the result with geographic context. Error handling should identify missing model features, invalid geometries, unavailable raster coverage and failed service responses rather than silently generating misleading results.

4.5 Standards and professional practice

The component should follow sound software-engineering practice: version control, documented interfaces, separation of concerns, input validation, repeatable testing and clear error reporting. Geospatial layers should retain their source, licensing and coordinate-reference metadata. IEEE referencing is used in this thesis in line with the template guidance. Security controls should follow the wider platform's authentication and role-based access design. Risk visualization should be accessible and consistent, with legendary and clear class labels. Predictions should be communicated as estimates and should not replace official warnings or professional judgment.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development of environment and technology stack

The wider project describes a web application using React.js, TypeScript, Tailwind CSS and Vite, with Leaflet/React-Leaflet for mapping and GeoJSON for spatial interchange. The backend uses Node.js and Express.js REST APIs, with WebSocket/Socket.IO for time-sensitive communication. Python and scikit-learn support machine-learning processing; the wider GIS service references pandas, GeoPandas, Folium, rasterio and Shapely for spatial data handling. MongoDB is identified as the shared data store. The exact versions and runtime configuration should be recorded from the actual development environment before final submission.

5.2 Component implementation

Component implementation connects spatial inputs, prediction results and the interactive frontend. The GIS processing stage reads the available input records and spatial layers, derives the required features where source data are available, and prepares the records expected by the trained model. Model output is associated with a location and risk category, then returned in a map-compatible format such as GeoJSON. The frontend renders the resulting features and provides geographic context. The implementation should ensure the prediction feature schema matches the training schema, and should log errors caused by missing files, invalid geometries or unavailable elevation coverage.

5.3 Integration with the other components

The flood map is Module 1 in the wider Flood Guard 360 architecture. It publishes risk information through the centralized integration layer. New water-level or rainfall observations from the monitoring module can be used to refresh risk information when the configured data flow supports recalculation. Flood-risk categories can help the rescue and relief module interpret locations alongside accessibility, affected populations and camp needs. This exchange is intended to avoid manual re-entry and provide a shared operational picture. The map does not independently control sensor devices, diagnose health conditions or allocate relief supplies.

5.4 Verification and software testing

Recommended verification includes: checking that valid input records are accepted; confirming that missing or malformed inputs produce clear errors; verifying coordinate systems and spatial alignment; checking that every prediction maps to an allowed risk class; confirming GeoJSON validity; testing that each risk class is rendered with the correct legend entry; confirming the map loads and supports inspection of geographic features; checking API responses and error handling; and verifying that updated risk output reaches the intended dashboard or integration endpoint. Model tests should use held-out data and report class-wise metrics. The supplied material does not provide a complete individual test log, so test execution evidence should be added from actual runs before submission.

5.5 Deployment and operation

In operation, the prediction service and web frontend should be configured with consistent model files, feature schemas and data locations. Input data should be refreshed according to their source update frequency, and each map layer should expose or record its update time. Monitoring should cover API availability, prediction failures, invalid spatial records and missing raster coverage. Backups and versioning should protect configuration and model artefacts. Before use in emergency operations, the component requires field validation, operational procedures and clear communication that model outputs are decision support rather than official flood warnings.

6 RESULTS AND VALIDATION

6.1 Research results

The flood-risk prediction component was evaluated using three machine-learning algorithms: Random Forest, Gradient Boosting, and Logistic Regression. Based on the reported experimental results, the Random Forest and Gradient Boosting models achieved an accuracy of 1.00 and an F1-score of 1.00 on the held-out test set. Logistic Regression achieved an accuracy of 0.93, an F1-score of 0.91, and an Area Under the Receiver Operating Characteristic Curve (AUC) of 0.95.

These results demonstrate the strong classification performance of the ensemble-based models within the evaluated dataset. Random Forest and Gradient Boosting performed better than the Logistic Regression baseline according to the reported accuracy and F1-score metrics. The AUC value obtained by Logistic Regression also indicates that the model demonstrated good discriminatory performance between the evaluated flood-risk classes.

The held-out test contained 40 observations. Although the reported results are promising, the relatively small test set should be considered when interpreting model performance and generalizability. Overall, the experimental findings support the suitability of machine-learning techniques for flood-risk classification within the scope of the project.

6.2 Product and technical results

The flood-map component successfully combines machine-learning-based flood-risk prediction with geographic information system (GIS) visualisation to present predicted risk levels in a spatially meaningful format. The system classifies flood risk into four categories: Low, Moderate, High, and Critical. These categories are represented using GIS-compatible spatial data and visualised through an interactive map developed using Leaflet and React-Leaflet.

The interactive map enables users to examine the geographic distribution of predicted flood risk, identify areas assigned different risk levels, and interpret prediction results in relation to their spatial context. The use of GeoJSON supports the structured

representation and exchange of geographic features, allowing predicted risk information to be consumed by other components of the Flood Guard 360 platform.

The technical implementation provides a foundation for presenting flood-risk information in a clear and accessible manner. The map interface connects the prediction output with its geographic representation, helping users understand the spatial implications of the model's classifications. The four-level risk categorisation also supports differentiated interpretation of predicted conditions, allowing areas with higher predicted risk to be distinguished from areas assigned lower risk.

Overall, the product results demonstrate the integration of predictive analytics and interactive mapping into a unified flood-risk visualisation component. This functionality supports the wider objective of Flood Guard 360 by making flood-risk predictions available in a form that can inform monitoring, emergency planning, and disaster-response activities.

6.3 User or stakeholder validation

The flood-map component was designed to support the needs of users who require clear and geographically relevant flood-risk information, including disaster-management personnel, emergency responders, and other stakeholders involved in flood preparedness and response. The interface presents predicted risk levels through distinct categories and an interactive geographic representation, enabling users to interpret spatial risk information and identify areas requiring further attention.

The map's design focuses on presenting risk information in an understandable format while preserving the geographic context of the predictions. The separation of risk into Low, Moderate, High, and Critical categories helps users distinguish between different levels of predicted risk. The interactive map also allows users to examine the spatial distribution of these classifications rather than relying solely on numerical model outputs.

From a usability perspective, the combination of geographic visualisation and clearly differentiated risk categories supports the interpretation of model predictions. The component is intended to reduce the difficulty of understanding spatial prediction

results and to provide a practical interface through which users can explore relevant areas.

The stakeholder-oriented design of the component contributes to its potential usefulness within a disaster-management environment. By presenting predicted risk information spatially, the map can support situational awareness and provide information that may assist subsequent planning and response decisions. Its operational effectiveness ultimately depends on the accuracy of the underlying predictions, the quality of geographic data, and the users' understanding of the displayed risk categories.

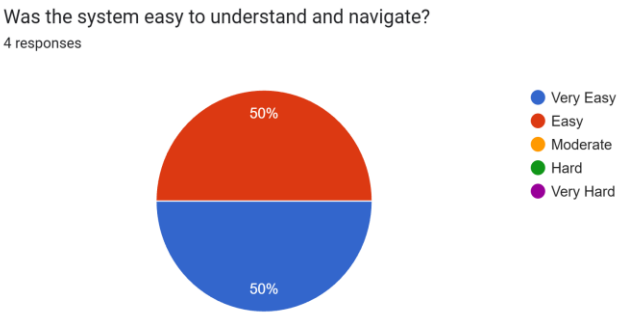


Figure 1

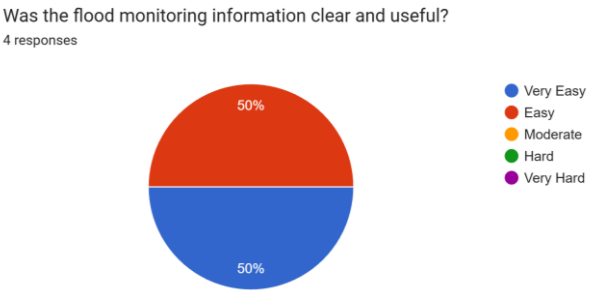


Figure 2

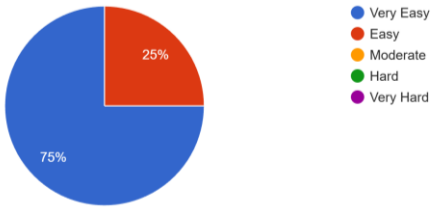


Figure 3

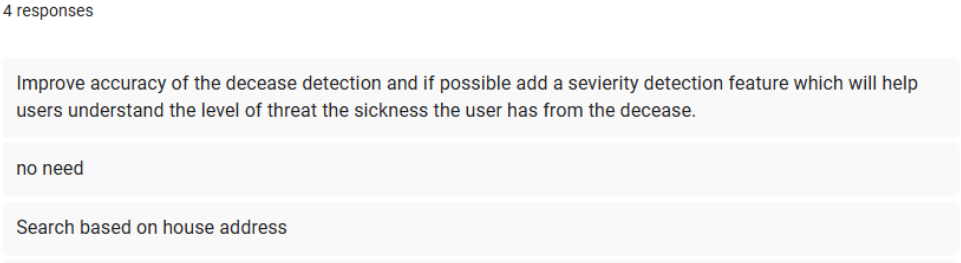


Figure 4

6.4 Integration results

The flood-map component is integrated into the wider Flood Guard 360 architecture to support the exchange and visualization of flood-risk information across the platform. The system architecture connects predictive analytics with geographic visualization and provides a shared mechanism for communicating relevant outputs to other components.

The flood-risk prediction module produces structured spatial information representing the predicted risk categories. This information is made available through the platform's shared API and data-exchange layer, allowing the map component to retrieve prediction results and display them geographically. GeoJSON provides a suitable format for representing spatial features and their associated risk attributes, supporting communication between the prediction service and the frontend application.

Integration with the wider platform enables flood-risk information to complement data generated by the river-monitoring component. Updated river-level observations can contribute to the flood-risk assessment process, while the resulting spatial predictions can provide additional context for rescue planning, evacuation-related decisions, and post-flood relief distribution. This establishes a connection between environmental monitoring, predictive analysis, and the presentation of risk information.

The integration architecture also supports the potential extension of the system through additional data sources and disaster-management modules. By separating prediction processing from geographic presentation and using a structured data-exchange mechanism, the component can be maintained and enhanced without requiring the entire platform to be redesigned.

Overall, the integration results demonstrate how the flood-map component fits within the wider Flood Guard 360 system and contribute to its objective of providing an integrated approach to flood-risk assessment and disaster response.

6.5 Summary of findings

The results demonstrate the potential of combining machine-learning-based flood-risk classification with GIS technology to produce an interactive and geographically meaningful flood-risk mapping solution. The reported experimental evaluation showed that Random Forest and Gradient Boosting achieved accuracy and F1-score values of 1.00, while Logistic Regression achieved an accuracy of 0.93, an F1-score of 0.91, and an AUC of 0.95. These findings indicate promising predictive performance within the evaluated dataset.

The developed flood-map component presents predicted risk information through four categories: Low, Moderate, High, and Critical. The integration of machine-learning outputs with GeoJSON and a Leaflet-based interactive map enables predictions to be interpreted in their geographic context. This approach provides a practical means of examining the spatial distribution of predicted flood risk and communicating the results to users.

The component also contributes to the wider Flood Guard 360 architecture by supporting the exchange of geospatial risk information between predictive analytics, environmental monitoring, and disaster-response functions. This integration establishes a technical foundation for using flood-risk predictions alongside other information relevant to emergency planning and relief operations.

In conclusion, the project demonstrates the value of combining predictive modelling, spatial data processing, and interactive visualisation within a unified flood-management platform. The findings support the feasibility of the proposed approach and establish a foundation for future improvements involving larger and more diverse datasets, enhanced geographic validation, continuous environmental data updates, and further evaluation under real-world flood conditions.

7 DISCUSSIONS

7.1 Interpretation of findings

The reported results suggest that the selected ensemble classifiers can separate the risk classes in the evaluation data used by the wider project. However, a perfect score on 40 observations may reflect the size or composition of the test set and is not sufficient evidence of performance across different districts, seasons or flood conditions. The GIS interface improves interpretability by associating categories with locations, but a clear map cannot compensate for incomplete or inaccurate input data. Both predictive validity and spatial correctness must be evaluated.

7.2 Comparison with literature baselines and existing solutions

The project design compares Random Forest and Gradient Boosting with Logistic Regression, providing two ensemble methods and a baseline within the same reported evaluation. The project paper reports higher metrics for the ensemble models on its test set, but the small sample size limits the strength of that comparison. Related literature supports the use of machine learning with GIS for flood-susceptibility assessment [1], [2], [3]. The available materials do not provide a standardized benchmark against commercial systems or an independently reproduced comparison, so broader claims of superiority are not justified.

7.3 Research contribution

The research contribution of this individual component is the applied design of a workflow that translates flood-risk model output into a spatial layer suitable for interactive inspection and integration. It brings together data preparation, classification output, geospatial serialization and browser-based visualization within the broader disaster-management architecture. The contribution is system integration and applied implementation rather than the invention of a new machine-learning algorithm.

7.4 Product contribution

The product contribution is a map interface intended to help users understand the geographic distribution of predicted flood risk. By showing risk categories in context and making outputs available to the shared platform, the component can support awareness and planning. Its usefulness depends on a readable legend, accurate geographic layers, timely data and transparent limitations. It should complement, not replace, official warnings and decisions by responsible authorities.

7.5 Limitations and threats to validity

The main limitations are the small, reported test set, possible gaps in labelled flood observations, dependency on the resolution and quality of elevation and other spatial layers, and uncertainty about generalization to locations not represented in training. Spatial leakage between training and test records could inflate metrics if nearby or duplicated records occur in both sets. Data-source differences, outdated rainfall or river-level inputs, invalid geometries and missing raster coverage can affect mapped outputs. The supplied materials do not document formal usability validation for this individual map. These limitations should be addressed through spatially independent evaluation, data-quality checks and field testing.

7.6 Implications

For disaster-management organizations, the component illustrates how predicted risk can be made easier to interpret through a spatial interface. For developers, it emphasizes the need to maintain consistent model features, spatial reference systems and output schemas across services. For researchers, it highlights that model metrics and map correctness are separate evaluation concerns. For public safety, the map should be used to support information with clear uncertainty communication and human oversight.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The Prediction-Based Flood Map component was developed as part of Flood Guard 360, an integrated flood-disaster management platform designed to support flood-risk assessment and disaster-response decision-making in Sri Lanka. The main purpose of this component is to identify and visualize flood-risk levels across different geographic locations by combining environmental data, spatial information, machine-learning techniques, and Geographic Information System (GIS) technologies.

The proposed component uses flood-conditioning factors such as rainfall, river-water level, elevation, land use, drainage, and soil-related information where available. The project research evaluates Random Forest and Gradient Boosting classifiers against Logistic Regression as a baseline model. The predicted results are categorized into four risk levels: Low, Moderate, High, and Critical. These classifications are represented as geospatial features and visualized through an interactive map using Leaflet-based mapping technologies. This enables users to interpret predicted flood risk in its geographic context rather than relying solely on numerical predictions or tabular information.

The reported evaluation in the wider project research paper showed that Random Forest and Gradient Boosting achieved accuracy and F1-scores of 1.00 on the stated held-out test set, while Logistic Regression achieved an accuracy of 0.93 and an AUC of 0.95. However, the evaluation used only 40 test observations. Therefore, the reported results should be interpreted cautiously and do not establish that the models will achieve the same performance across different districts, environmental conditions, or future flood events in Sri Lanka.

An important aspect of the component is its role within the wider Flood Guard 360 platform. Flood-risk information can be shared with the system's integration layer and used alongside real-time water-level monitoring, drainage alerts, rescue coordination, and relief distribution information. This integration provides a foundation for more coordinated flood-disaster management.

In conclusion, the Prediction-Based Flood Map demonstrates the potential of combining machine learning and GIS technologies to support location-based flood-risk assessment. Nevertheless, additional data collection, independent model validation, spatial-data verification, and real-world testing are necessary before the component can be considered reliable for operational disaster-response use.

8.2 Achievement of objectives

The objectives of the individual research component focused on developing a flood-risk prediction and visualization workflow that connects environmental data, machine-learning classification, and interactive geospatial mapping.

The first objective was to identify and prepare relevant spatial and environmental variables for flood-risk assessment. The project methodology identified rainfall, river-water level, elevation, land use, drainage, and soil-related information as relevant conditioning factors. The availability and quality of these inputs remain important considerations when applying the model to different geographic locations.

The second objective was to investigate suitable machine-learning approaches for flood-risk classification. The wider project research evaluated Random Forest and Gradient Boosting against Logistic Regression. The reported results favored the ensemble models on the available test set, although the small number of test observations limits the conclusions that can be drawn about generalization.

The third objective was to represent predicted flood-risk categories in a geographic format. The proposed workflow converts model predictions into a GIS-compatible output, such as GeoJSON, allowing each prediction to be associated with its corresponding geographic feature and risk category.

The fourth objective was to visualize the predicted flood-risk information through an interactive map. The project design uses Leaflet-based mapping technologies to display the predicted categories alongside relevant geographic features. This provides a more interpretable representation of the model outputs and allows users to examine the spatial distribution of flood risk.

The fifth objective was to support integration with the wider Flood Guard 360 platform. The component is designed to publish flood-risk information through the centralized integration layer so that monitoring and rescue or relief functions can use it as supporting information for operational decisions.

Overall, the project materials demonstrate the proposed methodology, architecture, model evaluation, and mapping workflow. However, the objectives relating to independent geographic validation, extensive live-system testing, and stakeholder evaluation require additional evidence before their full achievement can be confirmed.

8.3 Individual research and product contributions

The individual research contribution focuses on the Prediction-Based Flood Map component of Flood Guard 360. The work addresses the challenge of converting environmental and spatial information into geographically interpretable flood-risk classifications that can support disaster-management activities.

From a research perspective, the component brings together machine-learning classification and GIS-based visualization within a single workflow. It considers the use of Random Forest, Gradient Boosting, and Logistic Regression for flood-risk classification and examines how their outputs can be associated with geographic locations. The component also identifies the importance of spatial-data quality, model validation, and geographic generalization when developing flood-risk mapping systems.

From a product perspective, the contribution is the design of an interactive map that presents four flood-risk categories: Low, Moderate, High, and Critical. By combining predicted risk information with geographic context, the map is intended to help users identify locations that may require closer attention. The use of a GIS-compatible output also provides a practical mechanism for transferring model predictions between the prediction service and the web application.

Another important contribution is the component's integration within Flood Guard 360. Rather than functioning solely as an independent visualization tool, the flood map is designed to share its outputs with other system modules. This allows predicted flood-

risk information to complement live water-level observations, drainage-monitoring alerts, rescue planning, and relief-management activities.

The individual contribution should be distinguished from the wider group's work on IoT monitoring, post-flood health-risk screening, and relief-resource prioritization. The principal focus of this thesis is the design, analysis, and implementation requirements of the prediction-based flood-map function and its role within the integrated platform.

8.4 Future directions

Several improvements can strengthen the accuracy, reliability, and practical usefulness of the Prediction-Based Flood Map component.

Expansion of flood-related datasets: Future development should incorporate larger and more geographically diverse flood datasets covering different districts, rainfall patterns, seasons, land-use conditions, and flood severities. This would provide a stronger basis for training and evaluating the prediction models.

Improved geospatial data quality: Reliable elevation rasters, river-network datasets, drainage information, rainfall observations, and land-use data should be incorporated wherever available. The data-preparation workflow should validate coordinate reference systems, spatial alignment, geometry validity, missing values, and source-data resolution.

Independent model validation: Future evaluation should use larger test sets and, where possible, spatially or temporally independent observations. This would help determine whether the models generalize to locations and flood events that were not represented during training. Class-wise precision, recall, F1-scores, confusion matrices, and AUC should be examined to provide a more complete assessment of performance.

Integration with real-time monitoring: The component could be further developed to incorporate updated rainfall and river-water-level observations from the IoT monitoring module. This would support the refreshing of risk information as new environmental measurements become available, provided that the necessary data pipelines and prediction services are operational.

Enhanced map functionality: Future versions could improve layer controls, geographic search, location-specific risk details, update timestamps, and the display of relevant monitoring points. The interface should also communicate missing information and prediction uncertainty clearly so that users do not interpret model-generated classifications as guaranteed flood outcomes.

Field testing and stakeholder evaluation: The component should be tested against observed inundation records and evaluated with disaster-management personnel. Such testing could assess whether the map is understandable, whether risk categories are useful for operational planning, and whether the displayed information supports practical decision-making.

Reliability, security, and deployment improvements: Future work should include monitoring of prediction-service availability, error handling for missing spatial data, secure access to administrative features, and appropriate versioning of models and datasets. Deployment procedures should also ensure that risk information is updated consistently and that the map remains usable during periods of poor connectivity.

In summary, future development should prioritize reliable data, independent validation, real-world testing, and effective integration with the wider Flood Guard 360 platform. These improvements would help establish whether the Prediction-Based Flood Map can provide dependable decision-support information for flood-risk management in Sri Lanka.

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GLOSSARY

- GeoJSON (Geographic JavaScript Object Notation) – A JSON-based format for representing geographic features and their associated attributes.
- Geographic Information System (GIS) – A system used to collect, manage, analyze, and visualize geographically referenced data.
- Flood Risk Classification – The process of assigning geographic locations to risk categories based on relevant environmental and spatial factors.
- Flood Susceptibility – The relative likelihood that a geographic location may experience flooding based on its environmental and physical characteristics.
- Machine Learning (ML) – A branch of artificial intelligence that enables computer systems to identify patterns in data and generate predictions or classifications.
- Random Forest – An ensemble machine-learning algorithm that combines multiple decision trees to produce predictions or classifications.
- Gradient Boosting – An ensemble-learning technique that builds predictive models sequentially to improve performance by reducing previous prediction errors.
- Logistic Regression – A statistical classification algorithm used to estimate the probability of an observation belonging to a particular class.
- Leaflet – An open-source JavaScript library used to create interactive web maps.
- React-Leaflet – A library that provides React components for integrating Leaflet maps into React applications.
- Raster Data – Geographic data represented as a grid of cells or pixels, commonly used to represent elevation and other continuous environmental variables.
- Shapefile – A geospatial vector-data format used to store geographic features such as points, lines, and polygons.
- Spatial Data – Information that describes the geographic location, shape, or spatial relationships of features.
- Feature Engineering – The process of preparing, transforming, or deriving input variables to make them suitable for a machine-learning model.
- Model Validation – The process of assessing a model's performance using data that was not used to train it.
- Accuracy – The proportion of predictions that are correctly classified out of all evaluated predictions.
- F1-score – A performance measure calculated as the harmonic mean of precision and recall.
- Area Under the Curve (AUC) – A measure commonly used to evaluate how well a classification model distinguishes between classes across different decision thresholds.
- Application Programming Interface (API) – An interface that enables software components to communicate and exchange data.

- Internet of Things (IoT) – A network of connected physical devices that collect and exchange data through communication technologies.
- Flood Guard 360 – The integrated flood-disaster management platform that includes flood-risk mapping, IoT monitoring, post-flood health-risk screening, and rescue and relief management.

APPENDICES

Appendix A: Research Instruments and Extended Results

This appendix should provide supporting technical evidence for the Prediction-Based Flood Map component that is too detailed for the main chapters.

A.1 Flood-map input data

Document the datasets and spatial layers used by the component.

Data item	Purpose
Historical flood records	Provide information for flood-risk model development and evaluation
Rainfall data	Represent rainfall conditions associated with flooding
River-water-level data	Represent river conditions relevant to flood risk
Elevation data	Describe terrain characteristics
River-network data	Provide geographic information about rivers and support river-related feature engineering
Land-use data	Represent land-use characteristics relevant to flood susceptibility
Drainage data	Provide information about drainage conditions and infrastructure
Soil-related data	Represent soil characteristics where available

Table 1

For each actual dataset, include its source, geographic coverage, resolution, format, and preprocessing steps. Remove any items that were not used in your implementation.

A.2 Machine-learning evaluation results

The wider Flood Guard 360 research paper reports the following flood-risk model results.

Model	Accuracy	F1-score	AUC
Random Forest	1.00	1.00	1.00
Gradient Boosting	1.00	1.00	1.00
Logistic Regression	0.93	0.91	0.95

Table 2

Source: Flood Guard 360 group research paper. The reported held-out test set contains 40 observations; the results should therefore be interpreted cautiously.

Appendix B: Academic Integrity and AI Use Disclosure

Academic Integrity Declaration

I declare that this individual thesis, titled Prediction-Based Flood Map, has been prepared in accordance with the principles of academic integrity. All external sources, research publications, datasets, software tools, and other materials used in preparing this thesis must be appropriately acknowledged and referenced.

I acknowledge that I am responsible for the accuracy, originality, technical validity, and proper referencing of the submitted work. I understand that research findings, model performance results, and implementation details must accurately reflect the work undertaken.

AI Use Disclosure

Generative AI tools were used to assist with drafting, organizing, and refining selected sections of this thesis. The assistance included improving the clarity of technical explanations, structuring thesis content, and preparing descriptions of the Prediction-Based Flood Map component.

The research topic, individual component scope, and project-specific technical context were based on the Flood Guard 360 project materials. AI-generated content was reviewed for relevance and consistency with the available project documentation. The student remains responsible for verifying technical statements, model evaluation results, references, implementation details, and conclusions before submission.

AI assistance was not treated as a substitute for implementation, experimental testing, independent analysis, or academic source verification. Any code, datasets, results, and claims included in the final thesis must accurately reflect the work performed by the student and the research team.

AI Tools Used and Their Purposes

Tool: ChatGPT (OpenAI)

Purpose of use:

- Assisting with the organization and drafting of thesis chapters and subsections.
- Improving the clarity, grammar, structure, and academic tone of written content.
- Drafting explanations of the Prediction-Based Flood Map component, including its objectives, methodology, architecture, implementation, and integration within Flood Guard 360.
- Assisting with the presentation of model evaluation results and the discussion of research limitations.
- Preparing supporting text for the conclusion, future directions, glossary, and appendices.

ChatGPT was used as a writing and explanatory aid. It was not treated as an authoritative source of academic evidence, and AI-generated references or technical claims should be independently checked before submission.